

JANA HAGEN, CLS, N.D.

Business Informatics | Data Analytics | Data Science

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PROFESSIONAL SUMMARY

Data professional with 25+ years of experience spanning business informatics, data analytics, and data science across healthcare, clinical research, and small-business operations. Background includes designing and maintaining relational database and practice-management systems, delivering BI dashboards and reporting that drive operational decisions, and building machine learning and predictive models. Proficient in Python, SQL, DAX, and statistical/ML frameworks, with a track record of translating complex technical analysis into clear, actionable guidance for both technical and non-technical stakeholders. Completed MIT's Applied Data Science Program in November 2025 and actively applies ML/DL and generative AI skills through ongoing Kaggle and independent projects.

CORE TECHNICAL SKILLS

Business Informatics	Data Analytics	Data Science
- Relational database design (4D, SQL) - EHR & practice-management system design - Clinical coding data structures (ICD-9/10, CPT) - Electronic billing & revenue-cycle workflows - Business process & systems documentation - Git, GCP, Snowflake	- Power BI dashboard design & DAX - Excel-based financial & clinical reporting - KPI development and business reporting - A/B testing & regression analysis - Risk stratification & outcomes reporting - Data visualization (Matplotlib, Power BI)	- Python (Pandas, NumPy, SciPy) - TensorFlow, PyTorch, Scikit-learn - Predictive modeling & computer vision - Biostatistics & Bayesian methods - Generative AI / LLM-assisted analytics - Statistical modeling & experiment design

CERTIFICATIONS

- Applied Data Science Program: Leveraging AI for Effective Decision-Making (MIT)
- Physician, Naturopathic Medicine (N.D.)
- Snowflake SnowPro Core Certified

PROFESSIONAL EXPERIENCE

Applied Data Scientist / Business Informatics

2025 - 2026

ImagineAltions, LLC | Remote

- Served in a combined business informatics and applied data science capacity, using AI-assisted Python scripting to design, build, and automate complex analytics and informatics workflows.
- Developed and maintained data pipelines to prepare, transform, and structure datasets for downstream analytics and modeling.
- Applied AI-assisted development practices to accelerate script development, testing, and iteration for complex data engineering and analytics tasks.
- Partnered on informatics projects translating raw data into structured outputs suitable for reporting and decision support.

Crisis Responder

Mar 2023 - Mar 2024

988 National Crisis Hotline - La Frontera-EMPACT | Tempe, AZ

- Evaluated complex clinical risk factors by analyzing behavioral health data from EHR systems with Python-based analytics to improve risk stratification and caller dispositions, improving Call Center efficiency and quality to >95%, exceeding national standards.
- Executed clinical program evaluation through statistical data processing of clinical documentation, applying Bayesian statistical methods and SciPy to assess clinical quality metrics and achieve 100% compliance.
- Translated complex healthcare analytics and predictive modeling results for non-technical leadership, communicating technical findings to influence the delivery of AI-informed clinical services.

Medical Leave for Health Recovery

2020 - 2023

QA/QI Analyst / Manager / Medical Technologist

Feb 2014 - Mar 2020

VA Health Care System | Seattle & Phoenix

- Led regional healthcare analytics initiatives for the VISN 20 & 22 laboratory and Quality Improvement team, extracting actionable insights from heterogeneous diagnostic datasets via SQL and Power BI.

- Applied regression techniques, biostatistics, and scikit-learn to validate method performance data, ensuring 100% adherence to regulatory clinical program evaluation standards.
- Deployed continuous improvement methods to establish an Authorized Trainer Program, improving documentation quality and outcomes research data collection; nationally recommended as a standard of care.

4D Relational Database Architect / Business Informatics Analyst

2007 - 2014

Recovering Health, LLC | Seattle, WA

Concurrent with PhenoPath Laboratories, below

- Architected and maintained a 4D relational database system powering the practice's management software, converting the practice from paper patient charts to a fully electronic record and billing system.
- Designed data structures to capture ICD-9/ICD-10 diagnostic coding and CPT procedure coding for every patient encounter, feeding automated electronic billing workflows.
- Built SQL-based extraction and reporting processes across patient, clinical, and billing data for a private practice focused on addiction/relapse prevention and preventive primary care.
- Performed data analysis in Excel and DAX and designed business intelligence dashboards that supported effective, data-informed decision-making for practice operations and finances.

Medical Technologist (Lead and Special Projects)

Jun 1999 - Jan 2014

PhenoPath Laboratories PLLC | Seattle, WA

- Designed automated data extraction protocols using SQL to analyze large bodies of relational claims and clinical data for outcomes research and research validation.
- Supported Contract Research on Clinical Trials: cleaning data, developing test protocols, and reporting results to technical personnel and executive stakeholders.
- Formulated diagnostic methods by applying statistical modeling and evaluation research design across thousands of patient data points to validate longitudinal performance.
- Directed instrument validation through predictive modeling of quality metrics to ensure high-reliability diagnostic outcomes.
- Acted as knowledge architect, translating technical validation results into standardized operating procedures that reduced protocol development time for multidisciplinary clinical teams.

Aviation Electronics Technician

Sep 1984 - Aug 1994

United States Navy | Multiple Duty Stations

- Maintained and repaired complex avionics systems for multiple types of Naval aircraft. Discharged with honor.

EDUCATION

MIT Applied Data Science Program - Certificate, Applied Data Science, ML/DL, Recommendation Systems, Gen AI | Remote | Jul 2025 - Nov 2025

Bastyr University - Doctor of Naturopathic Medicine | Seattle, WA | 2000 - 2007

University of Washington - B.S., Medical Technology | Seattle, WA | 1997 - 1999

TECHNICAL PROJECTS

MIT Shinkansen Hackathon

MIT Applied Data Science Program - October 2025

Placed 4th out of 100+ participants, missing 3rd place by a margin of 0.0001, in a competitive applied data science hackathon.

Kaggriculture - Agriculture Simulation Competition

Kaggle

Developed an iterative simulation agent through systematic replay analysis of top competitors, disciplined single-hypothesis version testing, and batch statistical validation (multi-seed A/B comparison). Identified and resolved core simulation logic defects and progressively improved win rate across 70+ tested versions to reach a confirmed best-performing agent.

RSNA Knee MRI Detection Challenge

Kaggle

Built a machine learning pipeline targeting macro-average AUC above 0.90 across 12 knee findings, incorporating external pretraining datasets. Diagnosed significant weak-supervision failures in automated report labeling and built diagnostic tooling to validate label quality before model training.

AI Agent Security Red-Teaming Challenge

Kaggle

Applied archive-based search techniques with local LLM inference to identify a timing-window vulnerability in an AI guardrail system. Resolved GPU inference constraints and SDK issues to execute large-scale automated search campaigns.

Malaria Cell-Image Classification

MIT Applied Data Science Capstone

Built and evaluated a computer vision model for malaria cell-image classification, achieving over 98% across F1 score, precision, recall, accuracy, and ROC-AUC.